Roll No.....

MID SEMESTER EXAMINATION MARCH - 2025

Name of the Program: B.Tech. (AE) *Semester:* **IV**

Name of the Course: Introduction to Unmanned Aerial System *Course Code:* **TAS 411**

Time: 90 minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	**CO 2**
(a)	What are the recent trends in UAV technology, particularly in swarm technology and urban aerial mobility? Explain in detail.	
	OR	
(b)	Explain in Detail about various sensors used in UAVs. How does collision avoidance system enhance the safety of UAVs operation?	
Q2	(10 marks)	**CO2**
(a)	What do you understand by photogrammetry UAVs? Describe the workflow of photogrammetry using UAVs from flight planning to data processing.	
	OR	
(b)	Explain briefly the principle of LiDAR technology in UAVs. What is the key advantage of using LiDAR UAVs for topographic Mapping compared to traditional methods?	
Q3	(10 marks)	**CO 1**
(a)	How are UAVs classified based on size, and what is the typical application of each size category? Discuss the advantage and disadvantage of fixed-wing and rotary-wing UAVs.	
	OR	
(b)	Differentiate between automatic UAVs and autonomous UAVs with suitable examples.	
Q4	(10 marks)	**CO 3**
(a)	Explain the role of artificial intelligence (AI) and computer vision in enhancing UAV autonomy.	
	OR	
(b)	What do you mean by adaptive flight control system? What are the advantages and limitations of adaptive flight control systems in UAVs?	
Q5	(10 marks)	**CO 3**
(a)	Evaluate the impact of sensor fusion techniques (IMU, LiDAR, GNSS, and vision-based systems) on enhancing UAV stability and accuracy in adverse weather conditions.	

	OR	
(b)	What do you understand by mission planning of UAVs? Discuss about various components and key consideration of mission planning.	